

Introduction to Voice Recognition Module

1. Module Overview

CI1302 is a next-generation high-performance neural network intelligent voice chip developed by Chipintelli. It integrates Chipintelli's self-developed brain neural network processor BNPU V3 and CPU core, with a system main frequency of up to 220MHz. It features built-in 640KByte of SRAM, integrated PMU power management unit and RC oscillator, dual-channel high-performance low-power Audio Codec, and multiple peripheral control interfaces including UART, IIC, IIS, PWM, GPIO, and PDM. The chip requires only a small number of peripheral components such as resistors and capacitors to implement various smart voice product hardware solutions, offering excellent cost performance.

The CI1302 chip features a brain neural network processor core (BNPU) that supports offline NN acceleration computing and voice signal processing hardware acceleration. With a CPU main frequency of up to 220MHz, it enables offline far-field voice recognition and has 2MB of built-in FLASH storage.

2. Working Principle

This module uses a passphrase mode for wake-up. Users need to speak the configured wake word to activate the voice interaction module. After activation, voice recognition can be performed. The factory firmware has the default wake keyword set to "Xiao Ya Xiao Ya". If no voice is detected within 20 seconds, the module will enter sleep mode, and re-wake-up is required for the next use. The module supports wake word modification, command word modification, and custom phrase entries.

3. Precautions

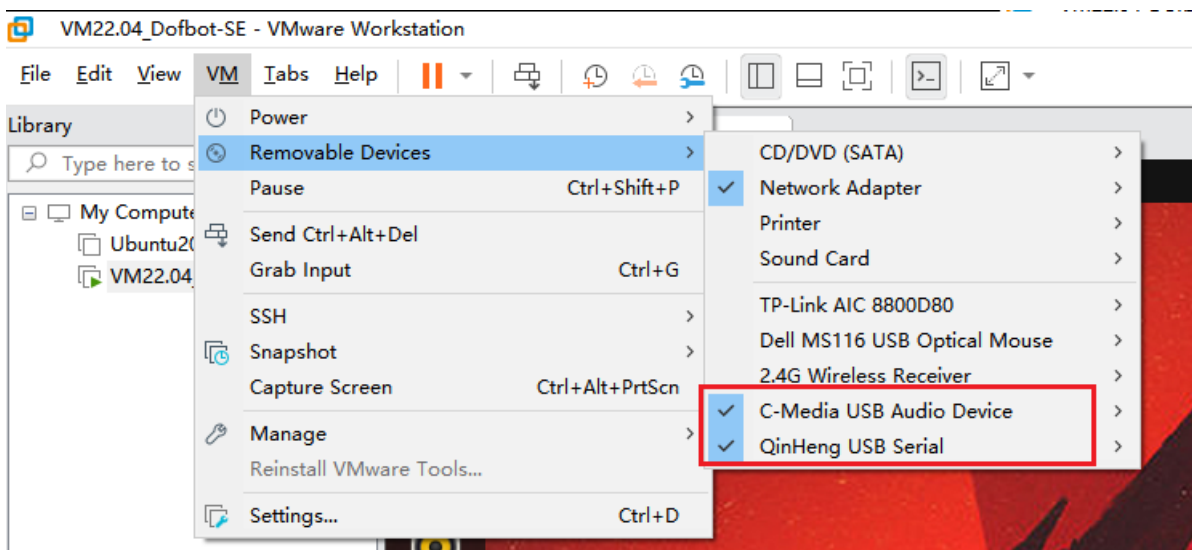
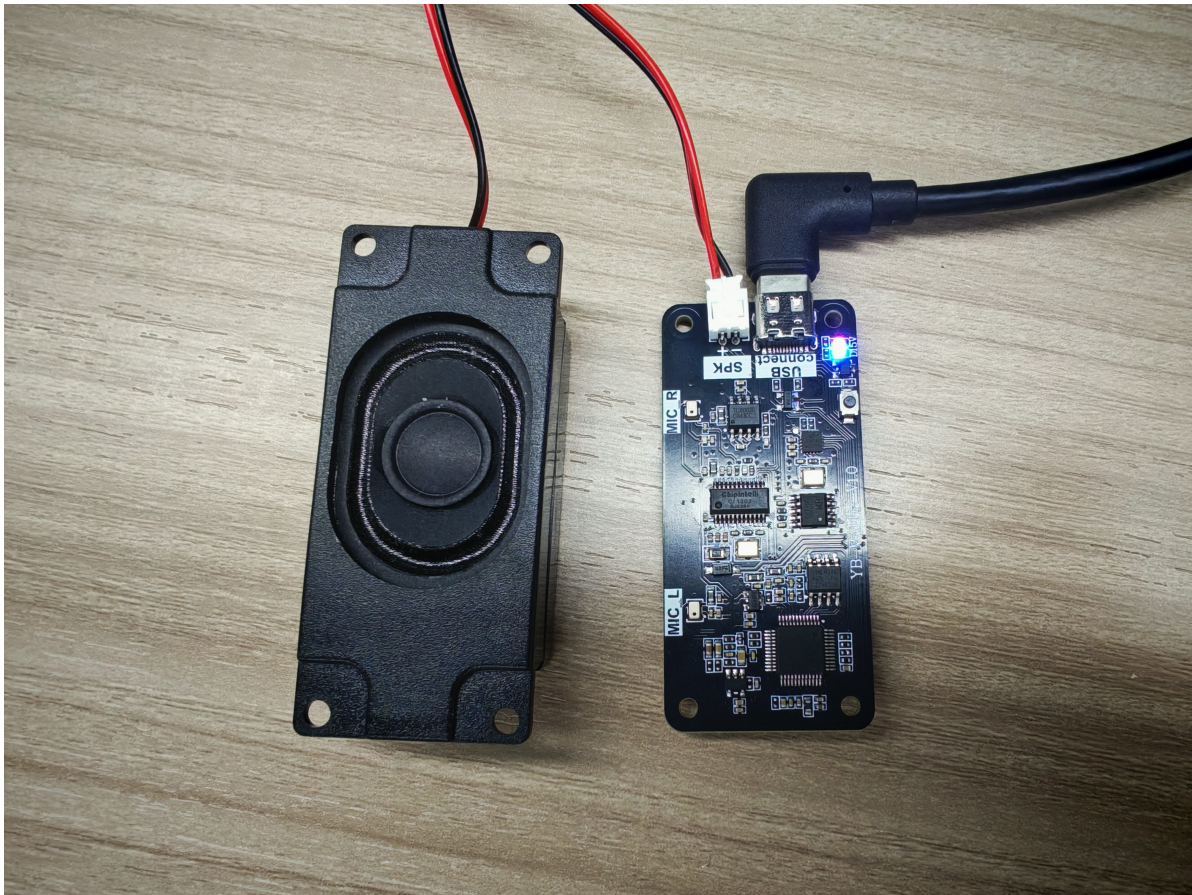
- Use 5V power supply; voltages exceeding 5V will damage the module
- The usage environment should be quiet; noisy environments will affect recognition performance
- When speaking phrases, your voice should be loud and the speaking speed should not be too fast. It is recommended to stay within 5 meters of the module

4. Driver Installation

The driver has been pre-installed in the provided virtual machine and can be used directly. This section explains the steps required to install the driver on a new motherboard or virtual machine. In the provided program source code folder, find the CH341SER folder and copy it to the motherboard directory. Open the terminal and enter the following commands for installation:

```
cd CH341SER
make -j6
sudo make install
sudo modprobe ch34x
```

After successful installation, connect the voice module to the virtual machine using a data cable, and the other end to the computer's USB port. It is recommended to prepare a USB HUB yourself.



Enter the following command:

```
11 /dev/ttyUSB*
```

If the following content is displayed, the installation is complete:

```
jetson@yahboom:~$ ls /dev/ttyUSB*  
/dev/ttyUSB0  
jetson@yahboom:~$
```

If you are using our provided virtual machine, enter the following command. If the display shows as follows, it indicates the voice module is properly connected to the virtual machine platform:

```
11 /dev/myspeech
```

```
yahboom@yahboom-virtual-machine: ~
yahboom@yahboom-virtual-machine: ~ 80x24
=====
IP : 192.168.2.32
DomainID : 30
=====
yahboom@yahboom-virtual-machine:~$ ll /dev/myspeech
lrwxrwxrwx 1 root root 7 Apr 23 11:37 /dev/myspeech -> ttyUSB0
yahboom@yahboom-virtual-machine:~$
```

5. Driver Library Installation

The Chinese driver library has been pre-installed in the provided virtual machine and can be used directly without reinstallation. This section explains the steps required to install the driver library on a new motherboard or virtual machine. In the provided program source code folder, find the `py_install_v0.0.3` folder and enter the following commands to install the driver library:

```
cd py_install_v0.0.3
sudo python3 setup.py install
```

After successful installation, you can use `pip3 list |grep Speech*` to verify the following content, indicating successful installation:

```
jetson@yahboom: ~$ pip3 list |grep Speech*
Speech-Lib 0.0.3
```

In the driver library, the default recognized port for the voice module is `/dev/ttyUSB0`. If the port is incorrect, you need to modify the content in the library and reinstall. The modification location is:

```
/py_install_v0.0.3/Speech_Lib/Speech_Lib.py
```

```
import time
import threading
import sys
import serial
import os

# V0.0.3
class Speech(object):
    def __init__(self, path = '~/speech_music/', com = '/dev/ttyUSB0'):
        # com="/dev/ttyUSB0"
        self.ser = serial.Serial(com, 115200)
        #self.ser = serial.Serial("/dev/ttyUSB0", 115200)
        self.language = 'cn' #默认是中文
        self.path = path

        if self.ser.isOpen():
            print("Speech Serial Opened! Baudrate=115200")
        else:
            print("Speech Serial Open Failed!")

    def __del__(self):
        self.ser.close()
        print("Speech Serial Closed!")
```

Change the value of `com` to the actual port value recognized by the voice module, which generally starts with `/dev/ttyUSB`. We map it to `/dev/myspeech`